

Handwriting Engine

You write a set of samples once with an Apple Pencil. After that you type anything, text or math, and get it back in your own handwriting, and you can put it on a worksheet. It all runs in the browser. Nothing is sent anywhere.

1. What you need

- An iPad with an Apple Pencil (a finger or mouse works, but looks worse).
- A computer with **Node 18 or newer** to host the page, *or* a GitHub account to publish your own copy (section 2).
- About 30 minutes to write the samples.

The code is public: `github.com/B0ngocat/Youtube-Views-Comment` (branch `handwriting-engine`). The handwriting is yours, so you teach it your own. Don't ask for someone else's samples file.

2. Get the app onto the iPad

The Pencil only works on the iPad itself, so the page has to open there. Pick one.

A. Same Wi-Fi (quickest)

```
git clone -b handwriting-engine https://github.com/B0ngocat/Youtube-Views-Comment.git
cd Youtube-Views-Comment
npm run serve
```

Then open `http://<your computer's address>:8080` in Safari on the iPad. The computer has to stay on while you use it.

B. Your own password-protected page (works anywhere, any time)

This builds one encrypted page and publishes it with GitHub Pages. Fork the repo first, so it goes to your account.

```
SITE_PASSWORD='a long password of 12+ characters' npm run deploy:pages
```

In your fork's settings, set Pages to deploy from the `gh-pages` branch. Wait for the "pages build and deployment" run to finish before you open the page (it can take a few minutes). Pick a long password: the encrypted file is public, so a short one can be guessed offline.

C. Someone gave you a link and a password

Open the link in Safari on the iPad and enter the password you were given. That's the whole setup: go straight to section 3. Your handwriting is saved in your own browser, not on the site, and nobody else

can see it. The **Download everything** button at the bottom left gives you a .zip with the code, the MCP server (section 6) and this guide, in case you want your own copy later.

Your samples are saved in the browser you teach in, on that web address only. A different address or browser starts empty. Use Export in the Teach tab to move them.

3. Teach it your handwriting

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Word 1 of 30

The quick brown fox jumps over the lazy dog.

the

Write the highlighted word on the solid line. Bigger and a little slower works better: try to fill the shaded band.

Next

☒ Ignore my palm (finger touches are ignored once the Pencil is used)

Coverage

LOWERCASE

a¹⁶ b⁷ c⁶ d¹² e²⁰ f⁵ g⁶ h⁸ i¹⁹ j⁴ k⁶ l⁹ m⁴ n¹¹ o¹⁹ p⁴ q⁵ r¹³ s⁹ t⁹ u¹²
v⁷ w⁷ x⁴ y¹⁰ z⁴

CAPITALS

A B¹ C D E F G H² I J K L M¹ N O P¹ Q R S² T² U V
W X Y Z

DIGITS

0 1 2 3 4 5 6 7 8 9

PUNCTUATION



The Teach tab on an iPad. (The writing in this picture is a test pattern, not a person.)

In the Teach tab, go through the rounds in this order. Each one asks for a word, letter or sentence at a time: write it on the solid line at normal speed and size, then tap Next.

1. **Alphabet, Capitals, Numbers, Symbols**
2. **Single letters:** each letter on its own. These are the cleanest examples it has.
3. **Math:** digits and symbols on their own (operators twice).
4. **Tricky letters:** a, e, o, r, u at the start, middle and end of words.
5. **Numbers in a row, Common words, Full lines**

Real words you write get reused as they are, so the Common words round makes every page look better.

Fixing wrong letters

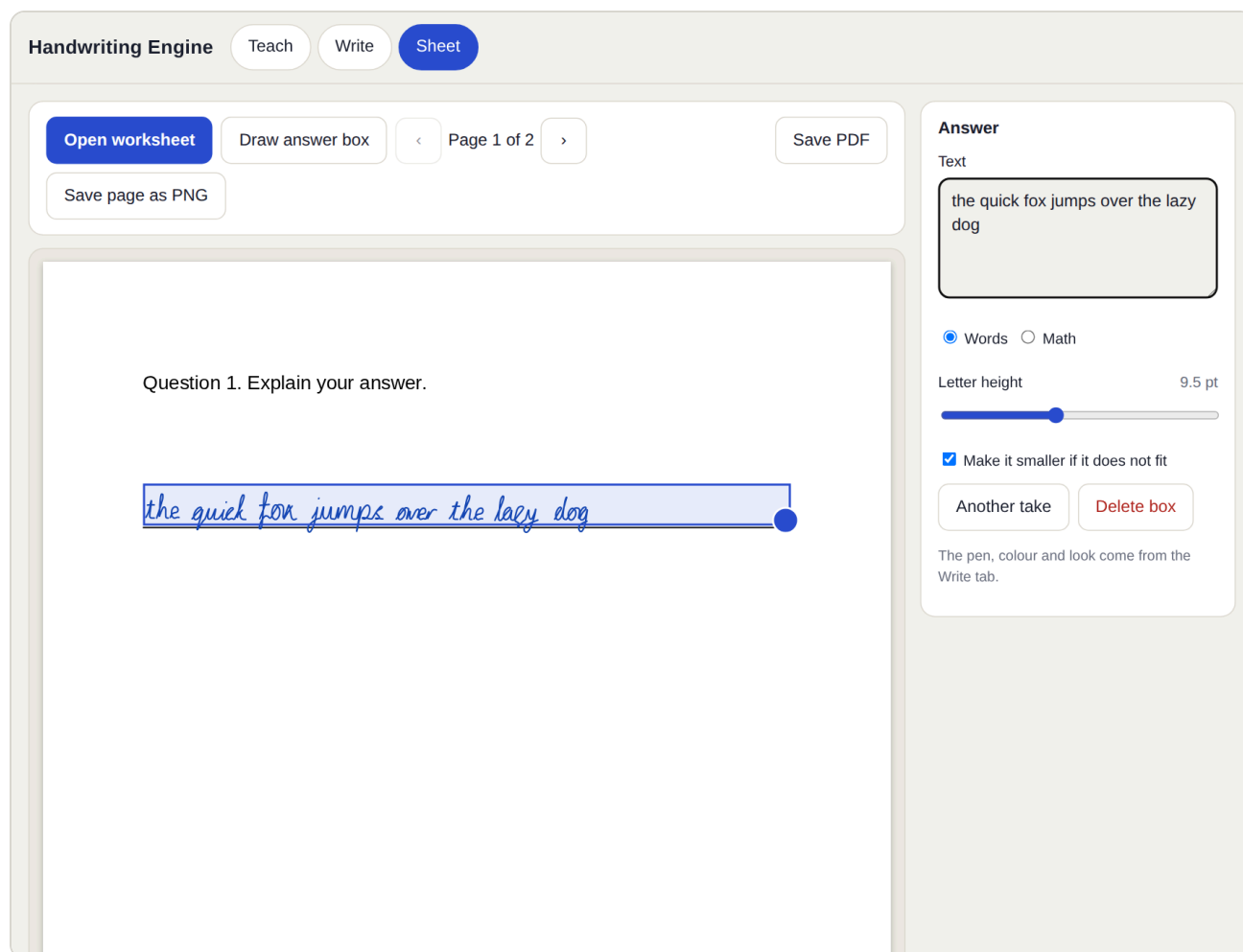
This is what improves quality most. Two ways:

- **Write tab, Fix mode:** tick it and tap any letter on the page that looks wrong. It is swapped for another example and the bad one is left out for good. Undo is right there.
- **Teach tab, "What it learned":** tap a letter in the Coverage grid to see every example it cut out for that letter. Tap the ones that don't look like the letter to leave them out. Do this for **a, e, o, r, u** and the digits first.

The sliders in the Write tab help too: **Neatness** (higher is easier to read) and **Use my real words**. If a result looks off, find the specific letters or words that give it away and fix those. "Make it more natural" doesn't do anything.

Export (Teach tab) saves `my-handwriting.json`. That file is your handwriting. Keep it private, don't post it, and back it up.

4. Put it on a worksheet (Sheet tab)



The Sheet tab: a box drawn on the page, the answer typed on the right, written in the user's hand.

1. **Open worksheet**: a PDF, or a photo or screenshot of one.
2. **Draw answer box**, then drag a rectangle where the answer goes. Drag a box to move it; use the round handle to resize it.
3. With a box selected, type the answer. It is written in your hand straight away. Choose **Math** for TeX-style input. **Another take** writes it differently.
4. An answer too long for its box is written smaller to fit (and says so).
5. **Save PDF** gives the worksheet back with the writing on it. **Save page as PNG** saves one page as a picture.

Boxes are remembered on that device for that file. A password-locked PDF has to be unlocked first. Pages rotated inside the PDF can be shown but not saved as PDF (use the PNG).

5. Math input

You type	You get
<code>x^2</code> , <code>x_1</code>	exponent, subscript
<code>\frac{a}{b}</code>	a stacked fraction
<code>\sqrt{x}</code> , <code>sqrt(x)</code> , <code>\sqrt[3]{x}</code> , <code>cubert(x)</code>	square and cube roots
<code>\int_0^1 x\,dx</code> , <code>\sum_{i=1}^n</code> , <code>\lim_{x \to 0}</code>	integrals, sums, limits
<code>\text{ so }</code>	plain words in the middle of math
<code>\</code> (backslash, space), <code>\quad</code>	a space that stays. Ordinary spaces are ignored, as in TeX.
<code>-></code> , <code><=</code> , <code>>=</code> , <code>!=</code>	$\rightarrow \leq \geq \neq$

A symbol you haven't written yet is drawn for you with a small wobble. It switches to your own once you write it in the Math round.

6. Letting an AI assistant use it (optional)

You can have an assistant such as Claude Code write in your hand and fill in PDFs for you. It runs on your computer from your exported file. Your handwriting never leaves your computer.

```
cd Youtube-Views-Comment
npm run build:mcp # makes dist/handwriting-mcp.js: one file, no
packages
claude mcp add handwriting -- node /full/path/dist/handwriting-mcp.js --samples
/full/path/my-handwriting.json
```

Add `--format svg` to get SVG back instead of PNG pictures. The tools: `write_text` , `write_batch` (many at once), `handwriting_status` . The PDF version, `node scripts/build-mcp.js --pdf` , adds `inspect_pdf` and `fill_pdf` .

Or paste this to an assistant that can run programs:

Use my handwriting server. My handwriting is in the file my-handwriting.json (keep it private: don't paste it anywhere or commit it). The server is handwriting-mcp.js, plain Node 18+, no npm packages. Run it over stdin/stdout:

```
node handwriting-mcp.js --samples my-handwriting.json --format svg
```

Tools:

- write_text {text, kind: "text"|"math", width_pt, letter_height_pt, ink, seed, format} format "svg" (SVG markup as text), "png" (a picture), or "both".
- write_batch {items: [...], format}: many at once. Use it instead of looping write_text.
- handwriting_status: which characters have no sample.
- inspect_pdf and fill_pdf (PDF build only): find where answers go on a PDF and write them in.

Math input: x^2 , $\frac{a}{b}$, $\sqrt{x}$ or `sqrt(x)`, $\sqrt[3]{x}$ or `cubert(x)`, $\int_0^1$, $\sum_{i=1}^n$, `\text{ words }`, and `"\ "` for a space that stays. The first call takes about 7 s; later starts take under half a second because it caches. Check every result before using it. Digits are the weakest part, and a missing character is reported, not guessed. Write only what I ask for. Don't fill in my name or anything else.

7. If something looks wrong

Problem	What to do
The page didn't update after publishing	GitHub's "pages build and deployment" run has to finish first. Then reload; browsers keep the old page for about 10 minutes. A home-screen app may need to be closed and reopened.
"Write" is empty or says 0 words	Samples are per browser and per web address. Teach there, or Import your exported file.
A letter keeps coming out wrong	Fix mode, or the Coverage grid (section 3). Add more of that letter in the Tricky letters round.
Numbers are hard to read	Do the Math round (digits on their own) and the Numbers in a row round, then use Fix mode on the digits.
Something on the page is cut off flat	The box is too small. Make it bigger, or the letters smaller.

One thing to decide for yourself. Whether it is okay to hand in handwriting made this way is up to you and your school's or workplace's rules. The tool doesn't decide that, and it never fills in your name or anything you didn't type.